National Teachers College
629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
School of Arts, Sciences, and Technology (SoAST)

SAGIPKABAYAN UN SDG 1 - No Poverty
A Digital Livelihood and Welfare Tracking System

A Software Design & Analysis Document

For the Final Project in

IT Elective 1

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1. INTRODUCTION

1.1 Project Overview

SagipKabayan is a localized digital livelihood management and welfare tracking platform engineered to assist Local Government Units (LGUs) and Public Employment Service Offices (PESO) in monitoring, managing, and mobilizing targeted employment interventions for highly vulnerable, digitally excluded, populations of low-income and homeless individuals. The platform effectively transitions manual, paper-heavy, and uncoordinated community tracking registries into a single centralized database environment.

The foundational concept of **SagipKabayan** is inspired by modern social media dynamics. On digital platforms, it is common for viral videos of elderly or severely marginalized homeless individuals begging for basic survival needs to spark public empathy. While internet viewers frequently try to coordinate and offer voluntary jobs to these individuals, digital comment sections are fleeting and disorganized. This system was built to formalize that community goodwill through a structured government framework.

Recognizing that homeless individuals and low-income residents suffer from extreme digital exclusion—often lacking smartphones, stable internet, or basic computer literacy—the system is designed to be hosted directly at the **LGU City Hall**. Instead of forcing vulnerable citizens to navigate a complex digital interface, they visit the City Hall where an LGU staff member assists them.

The application functions through a collaborative workflow:

1. **The Voluntary Employer:** An empathetic local citizen or business owner visits the City Hall and requests the administrator to post an available community service or temporary job opening (a **"Quest"**).
2. **The LGU Staff:** When a struggling resident or homeless individual seeks help, the LGU staff uses the interface to officially register them as an applicant beneficiary.
3. **The Matching:** The LGU staff views the open job listings and instantly maps the registered individual to the employer's voluntary posting, providing immediate employment and a vital economic lifeline.

Developed natively using the **VB.NET (Windows Forms) framework**, **ADO.NET data access layers**, and **Microsoft SQL Server**, the platform offers a secure, crash-resistant environment for administrators. The technical architecture enforces strict Object-Oriented Programming (OOP) principles, defensive input validation wrappers, safe data type conversions, parameterized database queries to eliminate SQL-injection vulnerabilities, and structural soft-deletion mechanics to maintain optimal relational integrity.

1.2 Chosen UN SDG Target

This project directly addresses **United Nations Sustainable Development Goal (UN SDG) 1 — No Poverty**, aiming to end poverty in all its structural forms through targeted local economic interventions. Specifically, the platform is mapped directly to two key target indicators:

- **SDG Target 1.1:** *By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than $2.15 a day.*
    - **System Alignment:** SagipKabayan acts as a direct financial lifeline tracker. It identifies extreme low-income earners and deploys them into paid community service tasks, ensuring clear visibility over immediate cash-for-work distributions.
- **SDG Target 1.4:** *By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance.*
    - **System Alignment:** The system utilizes digital technology to democratize access to temporary public work wages. By bridging the digital divide at the municipal level, it allows unbanked, marginalized individuals immediate access to structured municipal payouts via proxy staff interactions.

2. REQUIREMENTS ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operational behaviors, calculations, and processing tasks that the software system must execute.

Requirement ID	Requirement Name	Description

FR 1	User Authentication & Authorization	The system must validate user credentials (Username and Password) against tbl_Users and enforce role-based access restrictions on dashboard menus based on the assigned UserRole.
FR 2	Beneficiary Profile Management	The system must support complete CRUD lifecycles for marginalized citizens, capturing demographics, labor skill sets, and government verification documents into tbl_Employees.
FR 3	Soft-Delete Record Inactivation	The system must prevent permanent data loss by providing an inactivation mechanism that toggles the IsActive flag to 0 instead of executing hard SQL DELETE commands.
FR 4	Quest Ledger Management	The system must allow authorized administrators to post, edit, and audit municipal service projects, logging job details, specific financial rates (Gantimpala), and contact numbers into tbl_Quests.

FR 5	Strategic Assignment Mapping	The system must provide an active interface to map a registered resident to an open job posting, appending the worker's ID directly into the quest row and updating QuestStatus to 'Ongoing'.
FR 6	Activity History Reporting	The system must leverage joint multi-table database queries to compile historical deployment data and feed it directly into a professional Microsoft Report Viewer interface (frmReports.vb).

2.2 Non-Functional Requirements

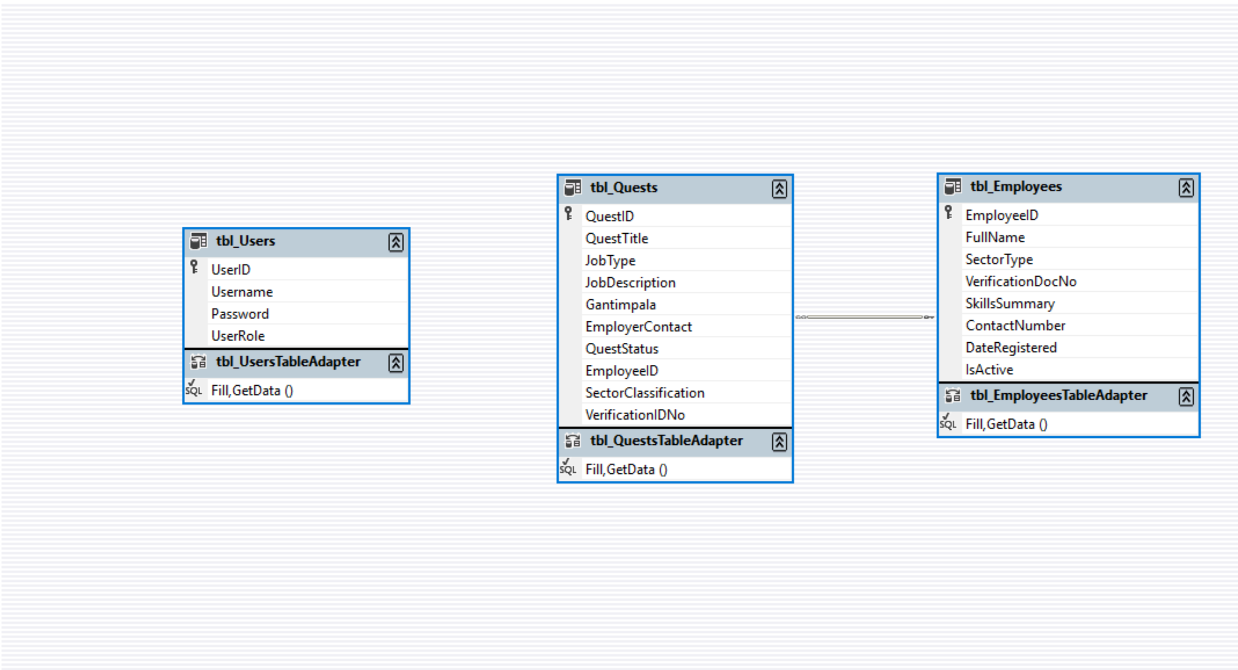
Non-functional requirements describe the structural constraints, quality attributes, and environmental performance parameters imposed on the runtime application.

Requirement ID	**Requirement Name**	**Quality Dimension**	**Description**
NFR 1	Parameterized Security	Security / Integrity	All SQL statements handling external data entries must utilize SqlParameter collections to block malicious SQL Injection attacks.

NFR 2	Data Type Defense	Reliability / Stability	The application must utilize defensive parsing (Integer.TryParse, Decimal.TryParse) to prevent runtime crashes caused by invalid data input conversions.
NFR 3	Form Navigation Modality	Usability / Workflow	Subforms (such as assignment and creation menus) must load as modal dialog boxes (ShowDialog()) to maintain structured, sequential administrative navigation tracking.
NFR 4	Relational Consistency	Fault Tolerance	The system architecture must mandate that no individual can be assigned to an active quest without a verified, preexisting primary key entry inside tbl_Employees.

3. DATABASE DESIGN

3.1 Entity Relationship Diagram (ERD)



3.2 Data Dictionary

Table 1: tbl_Users

This table manages administrative access, credentials, and structural system permission controls for LGU/PESO personnel who operate the application interface.

Field Name	Data Type	Constraints	Description
UserID	INT	Primary Key, Identity(1,1)	Unique auto-incrementing identifier for administrative accounts.

Username	VARCHAR(50)	NOT NULL	Alphanumeric handle used for system access authorization loops.
Password	VARCHAR(100)	NOT NULL	Secured credential string verified during login entry processes.
UserRole	VARCHAR(50)	NOT NULL	System clearance tier evaluating form permissions (e.g., Admin, Staff).

Table 2: tbl_Employees

This table maintains the localized digital profiles, skill summaries, and active registration states for the vulnerable and low-income community applicants.

Field Name	Data Type	Constraints	Description

EmployeeID	INT	Primary Key, Identity(1,1)	Unique tracking key assigned automatically to each registered citizen.
FullName	VARCHAR (100)	NOT NULL	The complete, legal name of the registered community member.
SectorType	VARCHAR (30)	NOT NULL	Demographic tracking classification segment (e.g., Homeless,

			Low-Income).
VerificationDocNo	VARCHAR (50)	NOT NULL	Reference tracking number mapped to an identity document (e.g., Certificate of Indigency or Social Welfare Case ID).
SkillsSummary	VARCHAR (100)	NOT NULL	Descriptive metadata detailing individual labor capacities

			(e.g., Heavy Labor, Janitorial, Repairs).
ContactNumber	VARCHAR(20)	NULL	Optional phone reference utilized for community deployment alerts.
DateRegistered	DATETIME	DEFAULT GETDATE()	Automatic timestamp recording when the worker profile was first entered

			into the LGU registry.
IsActive	INT	DEFAULT 1	Bit state tracking logical deletion flags (1 = Active, 0 = Soft-Deleted) used by the system's backend soft-delete routine.

Table 3: tbl_Quests

This table tracks available job openings, parameters, compensation rates, and the optimized caching properties connecting deployed applicants to jobs.

Field Name	Data Type	Constraints	Description
QuestID	INT	Primary Key, Identity(1, 1)	Unique auto-incrementing tracking identifier for community project listings.
QuestTitle	VARCHAR (100)	NOT NULL	Public designation title of the active job posting.

JobType	VARCHAR (20)	NOT NULL, DEFAULT 'Temporary'	Operational duration setting parameter (e.g., Temporary, Permanent).
JobDescription	TEXT	NOT NULL	Complete summary detailing specific project duties and safety guidelines.

Gantimpala	DECIMAL(10,2)	NOT NULL	Financial payout metrics or salary allocation for task completion. (Displayed as 'Salary' or 'Reward' in user interface forms and data grid rows for user-friendly accessibility).

EmployerContact	VARCHAR (20)	NOT NULL	Active contact phone reference for the hiring employer or voluntary entity.
QuestStatus	VARCHAR (20)	DEFAULT 'Available'	Deployment state tracker flag ('Available', 'Ongoing', or 'Completed').

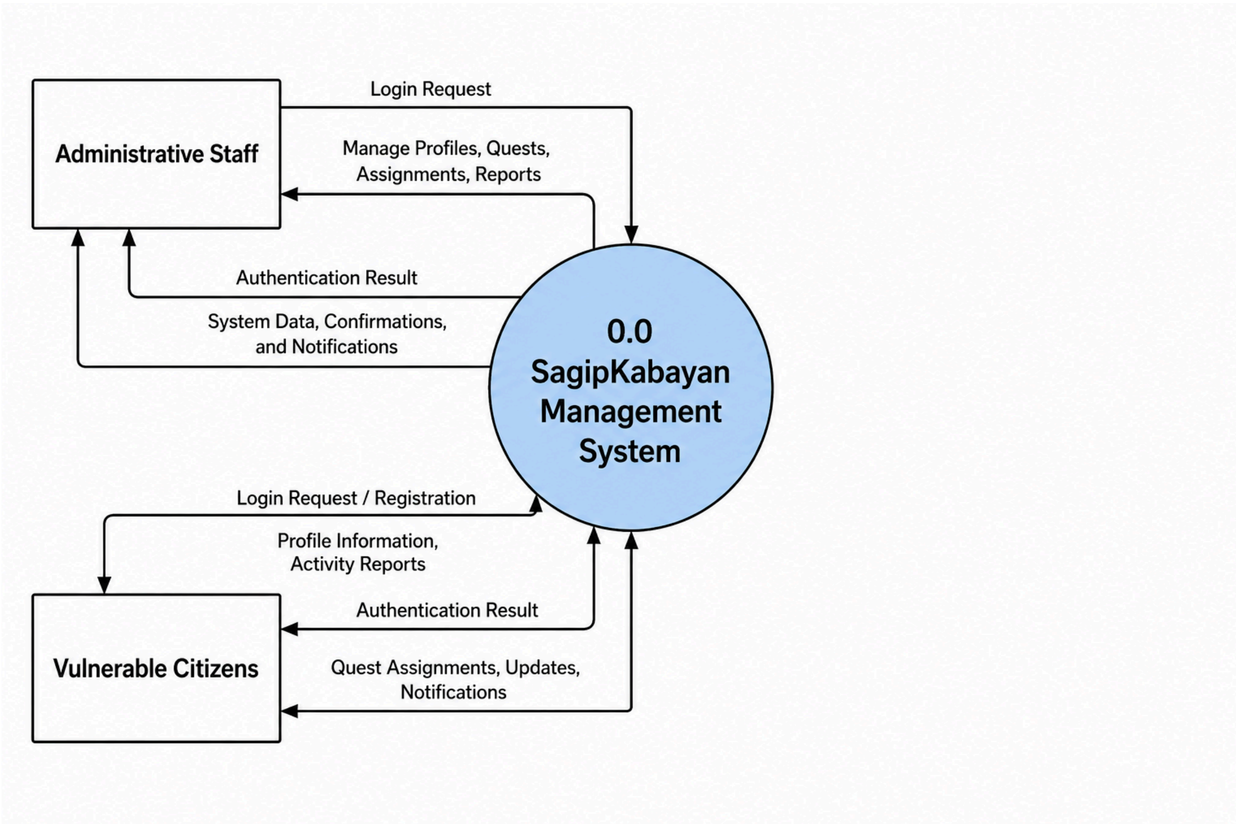
EmployeeID	**INT**	**Foreign Key -> tbl_Employees**	**Links a citizen to the job; remains completely empty or NULL if the job is unassigned.**
SectorClassification	**VARCHAR (100)**	**NULL**	**Optimized metadata caching column used by the system application layer to store and render demographic**

			insights quickly.
VerificationIDNo	VARCHAR (50)	NULL	Optimized metadata caching column reflecting the verified control ID number on the presentation layer forms.

4. SYSTEM ARCHITECTURE

4.1 Data Flow Diagram (DFD) Level 0 - Context Diagram

The Context Diagram establishes the high-level processing boundaries of the SagipKabayan platform, presenting the application as a single centralized system interacting with its external environments.



The scope is bounded by two primary operational interaction points:

- **Administrative Personnel (LGU / PESO Officers):** Act as the system operators. They feed input flows into the system boundary—including account login credentials, detailed beneficiary enrollment datasets, newly available community service postings, and active matching/deployment declarations. In response, the system boundary emits real-time output data streams, including system access confirmations, form processing state updates, updated data grid matrix views, and analytical output reports.
- **Vulnerable Citizens / Employers:** Act as the source foundation of raw data inputs. Citizen data profiles (demographics, labor capability summaries, and legal tracking document references) and job parameters (compensation quotas, duration limits, and point-of-contact details) flow into the administrative layer where they are logged directly into the system ecosystem.

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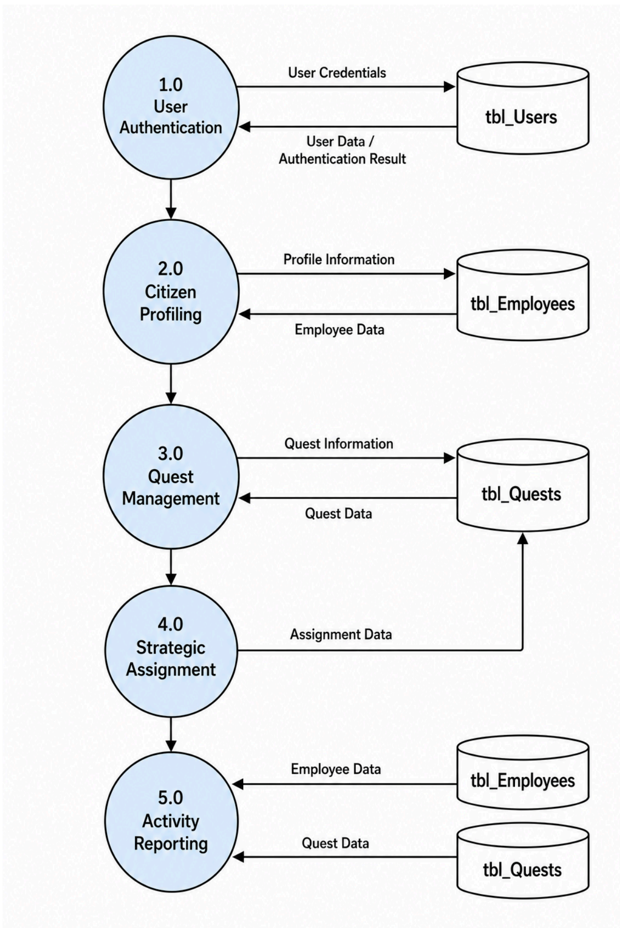
4.2 Data Flow Diagram (DFD) Level 1 - Process Breakdown

The Level 1 diagram provides an exploded functional view of the inner processing routines, tracing how datasets traverse functional data transformation processes and interact with the database tables.



The functional pipeline consists of five major sub-processes:

1. **1.0 User Authentication and Session Verification:** Validates incoming staff credentials against tbl_Users to unlock dashboard forms matching specialized role boundaries.
2. **2.0 Citizen Beneficiary Profiling and Inactivation:** Handles enrollment logs for homeless and low-income residents, piping data records straight into tbl_Employees and managing soft-delete state updates.

3. **3.0 Quest Management and Allocation Tracking:** Captures voluntary job openings brought in by local employers, storing financial values and descriptive parameters into tbl_Quests with a default status of 'Available'.
4. **4.0 Worker-to-Quest Strategic Assignment:** The critical matching node where an operator selects an active worker and binds their EmployeeID as a foreign key value inside the chosen quest row, toggling its operational status to 'Ongoing'.
5. **5.0 Consolidated Ledger and Activity Reporting:** Executes structured multi-table relational joins to compile historical records, outputting operational statistics to the Microsoft Report Viewer engine.

5. INDIVIDUAL CONTRIBUTIONS

The team structural assignments below trace the balanced technical module allocation executed across the application framework.

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Member's Name	**Module / Form Assigned**	**Contribution Description**
Maguidad, William		Frontend
Mahilum, Lovely		Documentation
Paildilan, Mark		Frontend and Backend
Tago, Irish Nicole		Documentation